

SKILL-02

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

CODE:-

```
srikar@LAPTOP-KHLR30TF: ~/os-lab/skill$ cat skill-02.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define MAX_INPUT 100

int main()
{
    char input[MAX_INPUT];
    int position;
    char ch;

    struct termios old_terminal;
    struct termios new_terminal;

    printf("=====\\n");
    printf("SIMPLE INTERACTIVE SHELL\\n");
    printf("=====\\n");
    printf("Type 'exit' to quit.\\n\\n");

    while (1)
    {
        position = 0;

        printf("myshell> ");
        fflush(stdout);

        /* Get current terminal settings */
        tcgetattr(STDIN_FILENO, &old_terminal);
        new_terminal = old_terminal;

        /* Disable canonical mode and echo */
        new_terminal.c_lflag &= ~(ICANON | ECHO);

        tcsetattr(STDIN_FILENO, TCSANOW, &new_terminal);

        while (1)
        {
            position = 0;

            printf("myshell> ");
            fflush(stdout);

            /* Get current terminal settings */
            tcgetattr(STDIN_FILENO, &old_terminal);
            new_terminal = old_terminal;

            /* Disable canonical mode and echo */
            new_terminal.c_lflag &= ~(ICANON | ECHO);

            tcsetattr(STDIN_FILENO, TCSANOW, &new_terminal);

            while (1)
            {
                ch = getchar();

                /* Enter key */
                if (ch == '\\n' || ch == '\\r')
                {
                    input[position] = '\\0';
                    printf("\\n");
                    break;
                }

                /* Backspace key */
                if (ch == 127 || ch == 8)
                {
                    if (position > 0)
                    {
                        position--;
                        printf("\\b \\b");
                        fflush(stdout);
                    }
                    continue;
                }

                /* Store normal characters */
                if (position < MAX_INPUT - 1)
                {
                    input[position] = ch;
                    printf("%c", ch);
                    position++;
                }
            }
        }
    }
}
```

```

        position--;
        printf("\b \b");
        fflush(stdout);
    }
    continue;
}

/* Store normal characters */
if (position < MAX_INPUT - 1)
{
    input[position] = ch;
    position++;

    putchar(ch);
    fflush(stdout);
}

/* Restore terminal settings */
tcsetattr(STDIN_FILENO, TCSANOW, &old_terminal);

/* Exit condition */
if (strcmp(input, "exit") == 0)
{
    printf("Shell terminated.\n");
    break;
}

/* Empty input */
if (strlen(input) == 0)
{
    continue;
}

/* Display entered command */
printf("You entered: %s\n", input);
}

return 0;
}
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$

```

OUTPUT:-

```

Processing triggers for libc-bin (2.43-1ubuntu2.5) ...
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$ nano skill-02.c
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$ gcc skill-02.c -o skill-02
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$ ./skill-02
=====
SIMPLE INTERACTIVE SHELL
=====
Type 'exit' to quit.

myshell> hello
You entered: hello
myshell> ls -a
You entered: ls -a
myshell> exit
Shell terminated.
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$ nano Makefile
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$ make
make: Nothing to be done for 'all'.
Venu@LAPTOP-KHLR30TF:~/os-lab/skill$ make run
./skill-02
=====
SIMPLE INTERACTIVE SHELL
=====
Type 'exit' to quit.

myshell> hello
You entered: hello
myshell> exit
Shell terminated.

```